SLA Class Thresholds Reference (Latency, Jitter, Loss)

KB-NET-1002 | NetCore SD-WAN 5.0 | Last updated: 2026-07-30 | Status: current

Summary

This reference defines the NetCore SD-WAN 5.0 SLA class thresholds for latency, jitter, and loss. It also summarizes the default app-to-class intent, BFD-based measurement method, 120-second polling behavior, and the three-interval evaluation multiplier used before an SLA breach is declared.

Applies to: NetCore SD-WAN 5.0, NetCore Orchestrator, NCX-1000 branch edge, NCX-4000 regional hub, NCX-9000 data center edge

Overview

NetCore SD-WAN 5.0 uses SLA classes to determine whether an application flow should remain on its current transport or be moved to a better compliant transport. Tunnels across MPLS, Internet broadband, and LTE/5G are monitored with BFD probes, and NetCore Orchestrator evaluates the collected latency, jitter, and loss values against the SLA class assigned to the application.

This article is the authoritative threshold reference for release 5.0. For SLA class configuration, see KB-NET-1001. For breach escalation, ticket creation, and automatic failover behavior, see KB-NET-1003.

SLA class threshold reference

The following thresholds apply to NetCore SD-WAN 5.0. A transport is compliant for a class only when the applicable latency, jitter, and loss metrics are within the listed limits.

NetCore SD-WAN 5.0 SLA class thresholds

SLA class	Latency threshold	Jitter threshold	Loss threshold	Typical application intent
Voice	<= 150 ms	<= 30 ms	<= 1%	Low-latency real-time audio traffic
Video	<= 200 ms	<= 40 ms	<= 2%	Interactive or streaming video traffic
Business-Critical	<= 250 ms	<= 50 ms	<= 3%	Enterprise applications that require predictable performance
Best-Effort	<= 400 ms	N/A	<= 5%	General traffic without a jitter requirement

Default app-to-class mapping reference

NetCore SD-WAN classifies application traffic into SLA classes through the app-aware routing policy distributed by NetCore Orchestrator. The default intent is to assign delay-sensitive traffic to stricter classes and leave non-sensitive traffic in Best-Effort unless a policy in Orchestrator overrides the class assignment.

Use KB-NET-1001 when changing these assignments. A changed mapping does not change the thresholds in the previous table; it only changes which SLA class is evaluated for that application.

Default application intent to SLA class mapping

Application intent	Default SLA class	Evaluation behavior
Real-time voice	Voice	Evaluated against latency <= 150 ms, jitter <= 30 ms, and loss <= 1%
Interactive video	Video	Evaluated against latency <= 200 ms, jitter <= 40 ms, and loss <= 2%
Business-critical enterprise traffic	Business-Critical	Evaluated against latency <= 250 ms, jitter <= 50 ms, and loss <= 3%

Application intent	Default SLA class	Evaluation behavior
General or unclassified traffic	Best-Effort	Evaluated against latency <= 400 ms and loss <= 5%; jitter is not evaluated

Measurement and evaluation method

NetCore SD-WAN 5.0 monitors overlay tunnels with BFD probes. App-route statistics are evaluated every 120 seconds. The multiplier is 3, which means NetCore Orchestrator declares an SLA breach only after three consecutive non-compliant app-route polling intervals, approximately six minutes.

Within each 120-second polling interval, the reported values represent the interval statistics collected from BFD-monitored tunnel measurements for the transport path. NetCore Orchestrator compares the reported latency, jitter, and loss values for that interval to the SLA class thresholds. A single non-compliant interval is not sufficient to declare a breach in release 5.0.

NetCore SD-WAN 5.0 SLA evaluation parameters

Parameter	Release 5.0 value	Operational meaning
Tunnel measurement method	BFD probes	Transport tunnel health and performance are monitored continuously by BFD-based measurements
App-route poll interval	120 seconds	SLA statistics are evaluated once per polling interval
Multiplier	3	Three consecutive non-compliant intervals are required before a breach is declared
Approximate breach declaration time	~6 minutes	120 seconds x 3 intervals
Breach response	Automatic failover to the best compliant transport	See KB-NET-1003 for escalation and auto-failover details

CLI reference for validating thresholds and live statistics

Use the following commands on an NCX edge to confirm the installed SLA class definitions and view current app-route measurements. The output below is representative for NetCore SD-WAN 5.0.

```
ncx-edge# show sdwan sla-class
SLA Class             Latency(ms)   Jitter(ms)   Loss(%)
-------------------------------------------------------
Voice                 150           30           1
Video                 200           40           2
Business-Critical     250           50           3
Best-Effort           400           n/a          5

ncx-edge# show sdwan app-route statistics
Timestamp UTC          App-Class           Transport           Lat(ms)  Jit(ms)  Loss(%)  State
-------------------------------------------------------------------------------------------------
2026-11-06 14:20:00    Voice               MPLS                42       6        0.1      compliant
2026-11-06 14:20:00    Video               Internet broadband  88       12       0.3      compliant
2026-11-06 14:20:00    Business-Critical   MPLS                61       9        0.2      compliant
2026-11-06 14:20:00    Best-Effort         LTE/5G              178      n/a      0.8      compliant

ncx-edge# show sdwan bfd sessions
Peer-System      Transport           Local-If       State  Last-Up UTC
-----------------------------------------------------------------------
hub-ncx4000-01   MPLS                ge0/0          up     2026-11-06 09:11:32
hub-ncx4000-01   Internet broadband  ge0/1          up     2026-11-06 09:11:36
hub-ncx4000-01   LTE/5G              cellular0/0    up     2026-11-06 09:12:04
```

Notes on interpretation

For Best-Effort, jitter is not applicable and is not used to determine compliance. A Best-Effort path can be compliant when latency is <= 400 ms and loss is <= 5%, regardless of the jitter value reported elsewhere in telemetry.

A metric equal to the threshold is compliant. For example, Voice latency of 150 ms, jitter of 30 ms, and loss of 1% are still within the Voice SLA class.

A breach requires consecutive non-compliant polling intervals for the selected application class. If a path is non-compliant for one interval and then compliant during the next interval, the release 5.0 multiplier sequence resets.

Release 4.2 used a superseded breach model: 600-second poll interval, multiplier 5, approximately 50 minutes to declare a breach, manual failover only, and e-mail notification only. Do not use release 4.2 timing when interpreting NetCore SD-WAN 5.0 SLA behavior.

Thresholds do not define alarm handling by themselves. Operational alarm codes used elsewhere in the NetCore SD-WAN domain include NCS-SLA-2001 SLA Violation Detected, NCS-SLA-2002 SLA Restored, NCS-BFD-3004 BFD Session Flap, NCS-BFD-3005 BFD Session Down, NCS-ZTP-4001 ZTP Redirect Failed, NCS-ZTP-4003 Device Certificate Validation Failed, NCS-QOS-5002 Queue Drop Threshold Exceeded, and NCS-ORC-6001 Orchestrator Control Connection Lost. For SLA breach escalation behavior, see KB-NET-1003.

Compliance examples

Class	Measured interval values	Compliant?	Reason
Voice	149 ms latency, 28 ms jitter, 0.9% loss	Yes	All metrics are within Voice thresholds
Voice	151 ms latency, 20 ms jitter, 0.5% loss	No	Latency exceeds 150 ms
Video	190 ms latency, 41 ms jitter, 1.0% loss	No	Jitter exceeds 40 ms
Business-Critical	240 ms latency, 45 ms jitter, 3.0% loss	Yes	Loss equal to 3% is within threshold
Best-Effort	350 ms latency, n/a jitter, 4.8% loss	Yes	Latency and loss are within Best-Effort thresholds
Best-Effort	401 ms latency, n/a jitter, 1.0% loss	No	Latency exceeds 400 ms

Related Articles

KB-NET-1001: Configuring App-Aware Routing SLA Classes

KB-NET-1003: SLA Breach Policy: Escalation and Auto-Failover